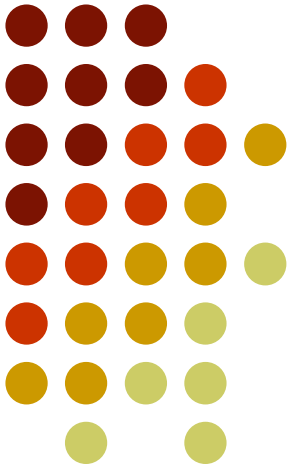
Mechanics of Options Markets

Chapter 9



Types of Options



- A call is an option to buy
- A put is an option to sell
- A European option can be exercised only at the end of its life
- An American option can be exercised at any time

Option Positions

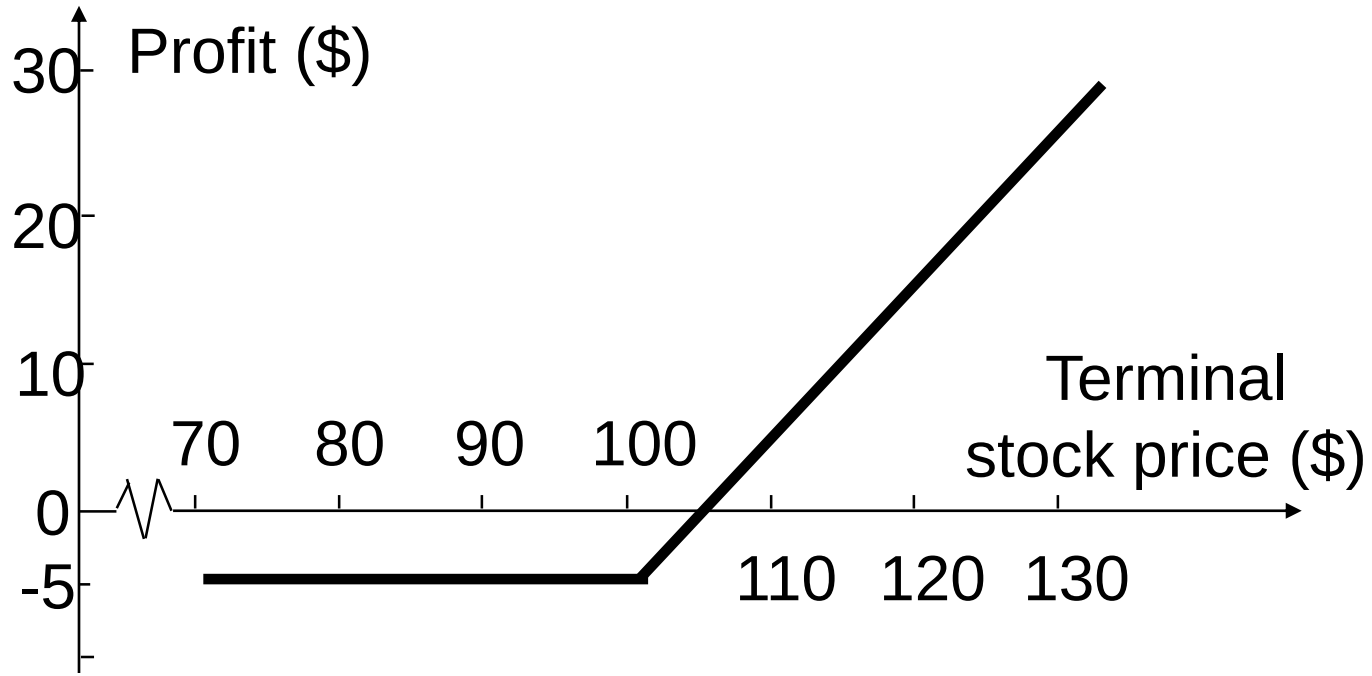


- Long call
- Long put
- Short call
- Short put

Long Call

(Figure 9.1, Page 212)

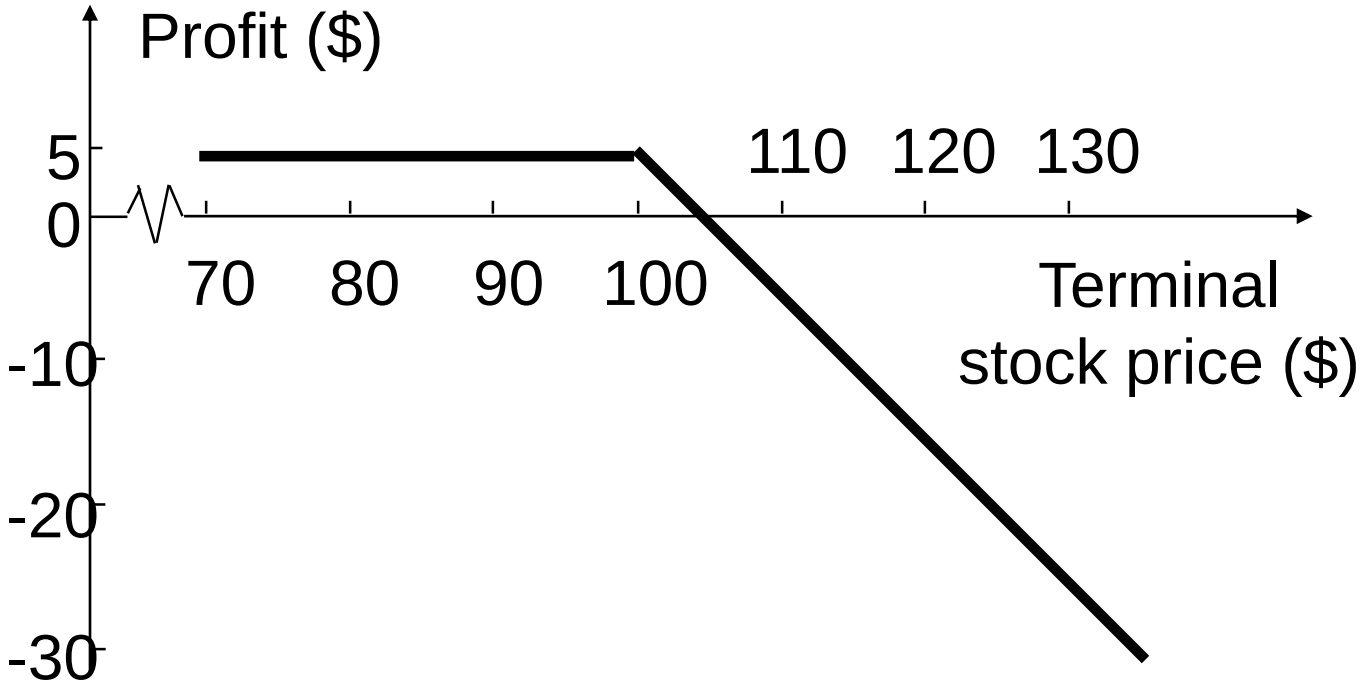
Profit from buying one European call option: option price = $5, strike price = $100.



Short Call

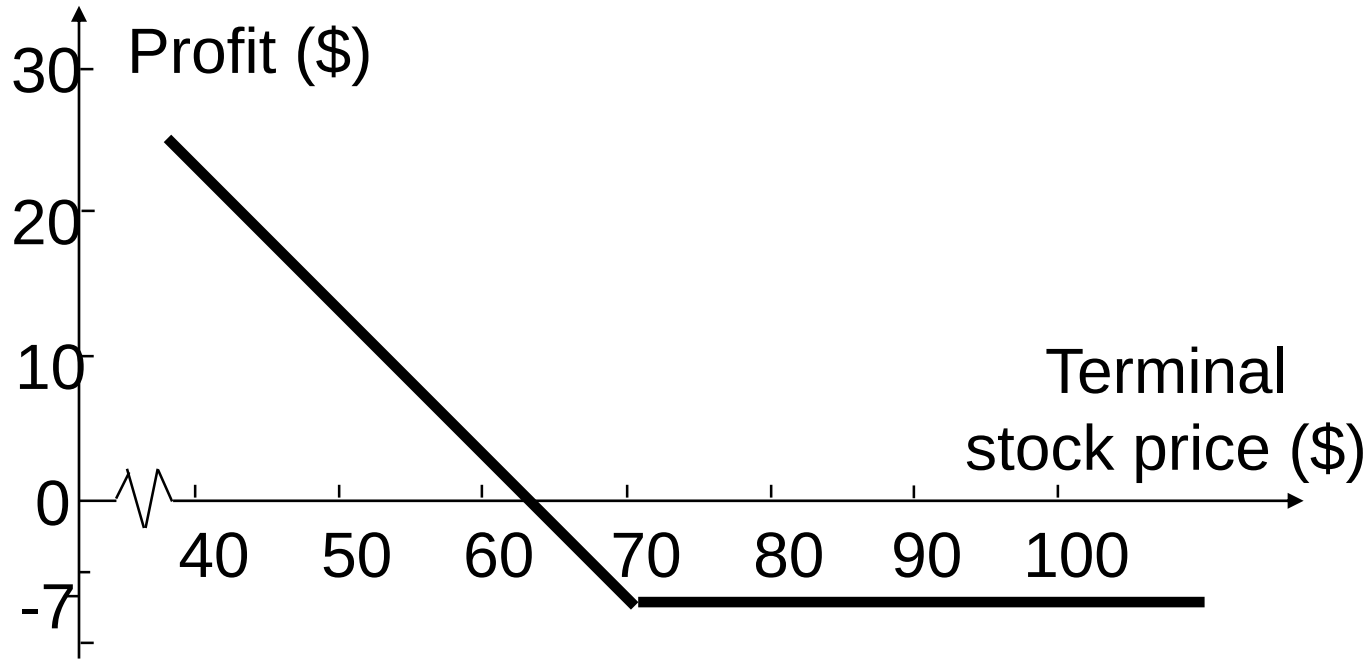
(Figure 9.3, page 213)

Profit from writing one European call option: option price = $5, strike price = $100



Long Put
(Figure 9.2, page 213)

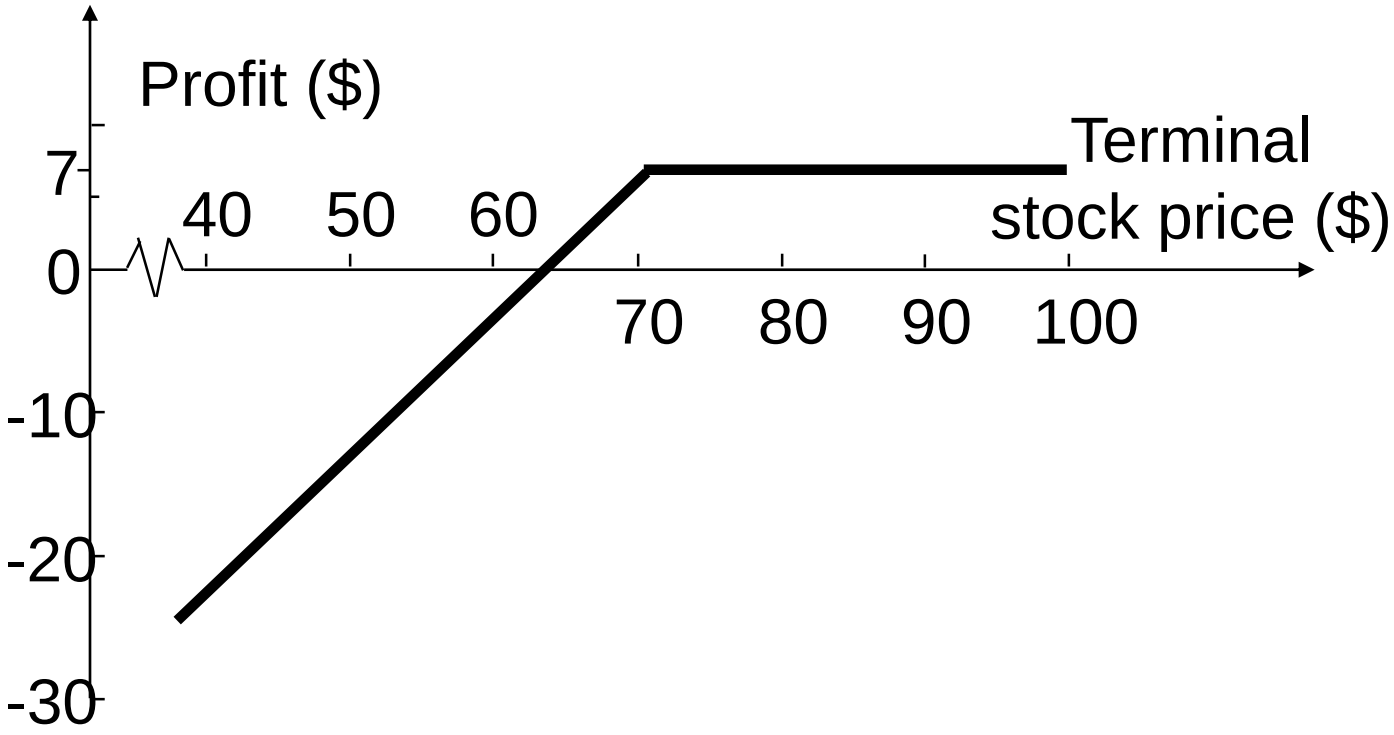
Profit from buying a European put option: option price = $7, strike price = $70



Short Put

(Figure 9.4, page 214)

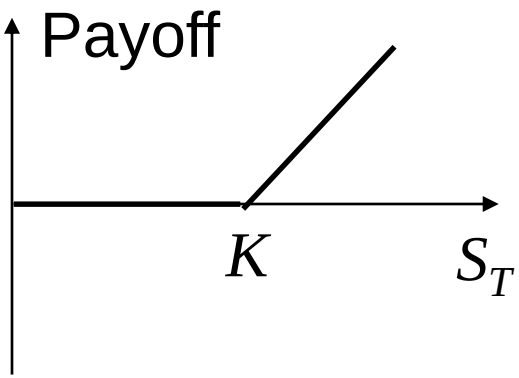
Profit from writing a European put option: option price = $7, strike price = $70

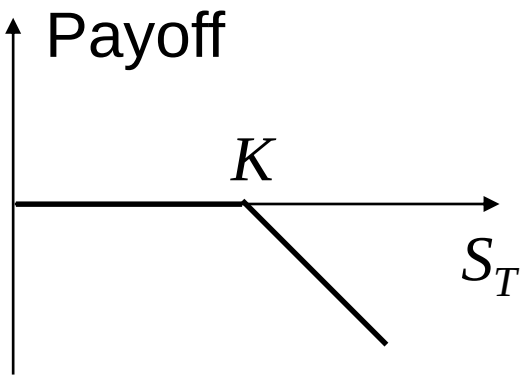


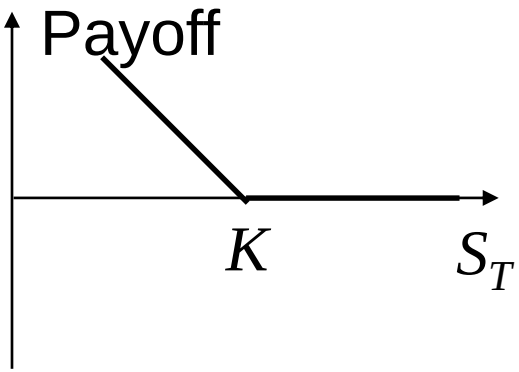
Payoffs from Options

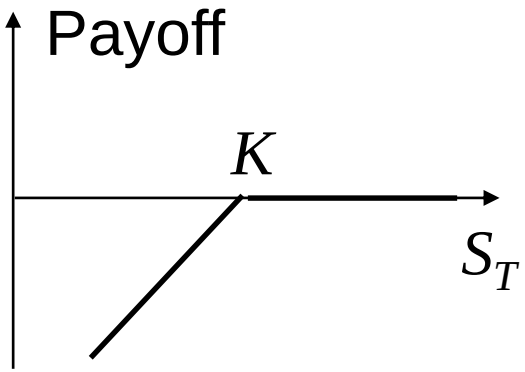
What is the Option Position in Each Case?

K = Strike price, S_T = Price of asset at maturity









Assets Underlying Exchange-Traded Options

Page 215-216



- Stocks
- Foreign Currency
- Stock Indices
- Futures

Specification of Exchange-Traded Options



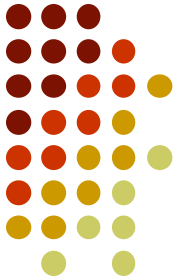
- Expiration date
- Strike price
- European or American
- Call or Put (option class)

Terminology

Moneyness :

- At-the-money option
- In-the-money option
- Out-of-the-money option

Terminology
(continued)



- Option class
- Option series
- Intrinsic value
- Time value

Dividends & Stock Splits

(Page 219-220)



- Suppose you own options with a strike price of K to buy (or sell) N shares:
  - No adjustments are made to the option terms for cash dividends
  - When there is an n-for-m stock split,
    - the strike price is reduced to mK/n
    - the number of shares that can be bought (or sold) is increased to nN/m
  - Stock dividends are handled in a manner similar to stock splits

Dividends & Stock Splits
(continued)

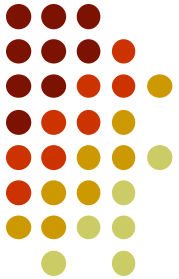


- Consider a call option to buy 100 shares for $20 per share
- How should terms be adjusted:
  - for a 2-for-1 stock split?
  - for a 25% stock dividend?

Market Makers

- Most exchanges use market makers to facilitate options trading
- A market maker quotes both bid and ask prices when requested
- The market maker does not know whether the individual requesting the quotes wants to buy or sell

Margin (Page 222-224)



- Margin is required when options are sold
- A naked option is an option that is not combined with an offsetting position in the underlying stock.
- When a naked call option is written in the US, the margin is the greater of:
  1. A total of 100% of the proceeds of the sale plus 20% of the underlying share price less the amount (if any) by which the option is out of the money
  2. A total of 100% of the proceeds of the sale plus 10% of the underlying share price

Margin (continued)



- When a naked put option is written in the US, the margin is the greater of:
  1. A total of 100% of the proceeds of the sale plus 20% of the underlying share price less the amount (if any) by which the option is out of the money
  2. A total of 100% of the proceeds of the sale plus 10% of the exercise price
- The 20% in the calculations is replaced by 15% for options on a broadly based stock index.

Example: Margin Calculations



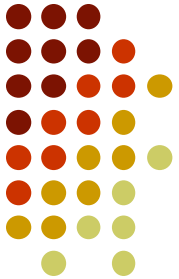
- An investor writes four naked call option contracts on a stock. The option price is $5, the strike price is $40, and the stock price is $38.
  - The option is $2 out of the money
  - First calculation: 400[5 + 0.2(38) – 2] = $4,240
  - Second calculation: 400[5 + 0.1(38)] = 3,520
- If the option had been a put, it would be $2 in the money. The margin requirement would be

  400[5 + 0.2(38)] = $5,040

Warrants

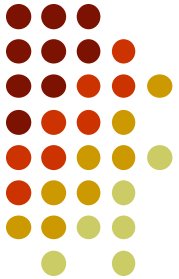
- Warrants are options that are issued by a corporation or a financial institution
- The number of warrants outstanding is determined by the size of the original issue and changes only when they are exercised or when they expire

Warrants
(continued)



- The issuer settles up with the holder when a warrant is exercised
- When call warrants are issued by a corporation on its own stock, exercise will usually lead to new stock being issued

Employee Stock Options
(see also Chapter 14)



- Employee stock options are a form of remuneration issued by a company to its employees
- They are usually at the money when issued
- When options are exercised the company issues more stock and sells it to the option holder for the strike price

Convertible Bonds

- Convertible bonds are regular bonds that can be exchanged for equity at certain times in the future according to a predetermined exchange ratio
- They are bonds with an embedded call option on the company's stock
- Usually a convertible is callable
- The call provision is a way in which the issuer can force conversion at a time earlier than the holder might otherwise choose

Summary

- Options types
  - Call and put
  - European option and American option
- Profit and payoff of different option positions
  - Long call, long put, short call, short put
- Terminology
  - Moneyness: at the money, in the money, out of the money
  - Option class, option series, intrinsic value, time value

Summary

- Impact of dividends and stock splits on option terms
  - Cash dividends: no change
  - Stock split and stock dividends: strike price reduces, number of shares increases
- Margin calculations
  - Margin is required when options are sold
  - Margin for naked call and naked put
  - Options on stock index: 15% is used instead of 20%